

EE 66
Fall 2026

Signals, Dynamics, and Information

Discussion 6A

1. Predicting with Least Squares

A courier would like to predict the time of a future delivery from its distance. The distance d_i of each delivery is known ahead of time from the route. The time t_i is what the courier measures, by timing four deliveries:

distance d_i (km)	1	2	3	4
time t_i (min)	6	5	10	11

$$\mathbf{d} = \begin{bmatrix} 1 \\ 2 \\ 3 \\ 4 \end{bmatrix}, \quad \mathbf{t} = \begin{bmatrix} 6 \\ 5 \\ 10 \\ 11 \end{bmatrix}.$$

- (a) First consider the model $t_i \approx s d_i$, in which time is proportional to distance. The slope s is the only unknown parameter. Write the four equations in the form $A\mathbf{x} \approx \mathbf{b}$, identifying A , $\mathbf{x}$ and $\mathbf{b}$. Find the least squares solution $\hat{\mathbf{x}}$ and compute the squared error $\|\mathbf{b} - A\hat{\mathbf{x}}\|^2$. What time does this model predict for a 5 km delivery?

- (b) Now consider the model $t_i \approx s d_i + c$, which adds an intercept c . Again write the four equations in the form $A\mathbf{x} \approx \mathbf{b}$, now with $\mathbf{x} = \begin{bmatrix} s & c \end{bmatrix}^\top$. Find the least squares solution $\hat{\mathbf{x}}$, compute the squared error $\|\mathbf{b} - A\hat{\mathbf{x}}\|^2$, and predict the time of a 5 km delivery.

Compare the squared error with part (a). Which model fits the observations better, and why? Could adding the intercept ever have made the squared error larger?

2. Least Squares: How A Shapes the Answer

Least squares finds the prediction $\mathbf{A}\hat{\mathbf{x}}$ in $\text{Col}(\mathbf{A})$ that is closest to $\mathbf{b}$. That prediction is the **projection** of $\mathbf{b}$ onto the subspace $\text{Col}(\mathbf{A})$, and the residual $\mathbf{r} = \mathbf{b} - \mathbf{A}\hat{\mathbf{x}}$ is perpendicular to every column of $\mathbf{A}$:

$$\mathbf{A}^\top \mathbf{r} = \mathbf{0}.$$

In this problem the target stays fixed while the matrix changes:

$$\mathbf{b} = \begin{bmatrix} 1 \\ 3 \\ 4 \end{bmatrix}.$$

Very little computation is needed. Try to answer each part from the geometry before computing anything.

(a) Let

$$\mathbf{A}_a = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{bmatrix}.$$

Describe $\text{Col}(\mathbf{A}_a)$ in words. Without using the formula $\hat{\mathbf{x}} = (\mathbf{A}^\top \mathbf{A})^{-1} \mathbf{A}^\top \mathbf{b}$, find the projection $\mathbf{A}_a \hat{\mathbf{x}}$, the residual $\mathbf{r} = \mathbf{b} - \mathbf{A}_a \hat{\mathbf{x}}$, the squared error $\|\mathbf{r}\|^2$, and the parameters $\hat{\mathbf{x}}$. Then verify that $\mathbf{A}_a^\top \mathbf{r} = \mathbf{0}$.

(b) Now let

$$\mathbf{A}_b = \begin{bmatrix} 1 & 1 \\ 1 & -1 \\ 0 & 0 \end{bmatrix}.$$

How does $\text{Col}(\mathbf{A}_b)$ compare with $\text{Col}(\mathbf{A}_a)$? Which of the projection $\mathbf{A}\hat{\mathbf{x}}$, the residual $\mathbf{r}$, the squared error $\|\mathbf{r}\|^2$ and the parameters $\hat{\mathbf{x}}$ change from part (a), and which do not? Find the new $\hat{\mathbf{x}}$.

(c) Now let

$$\mathbf{A}_c = \begin{bmatrix} 1 & 2 \\ 1 & 2 \\ 0 & 0 \end{bmatrix}.$$

- i. Describe $\text{Col}(\mathbf{A}_c)$ and find the projection $\mathbf{A}_c \hat{\mathbf{x}}$, the residual $\mathbf{r} = \mathbf{b} - \mathbf{A}_c \hat{\mathbf{x}}$ and $\|\mathbf{r}\|^2$.
- ii. Before computing, could the squared error have come out smaller than in part (a)? Why or why not?
- iii. Compute $\mathbf{A}_c^\top \mathbf{A}_c$. What goes wrong with the formula $\hat{\mathbf{x}} = (\mathbf{A}^\top \mathbf{A})^{-1} \mathbf{A}^\top \mathbf{b}$? Find every $\hat{\mathbf{x}}$ for which $\mathbf{A}_c \hat{\mathbf{x}}$ equals the projection you found above.

(d) Finally, let

$$\mathbf{A}_d = \begin{bmatrix} 1 & 1 & 0 \\ 1 & -1 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

What is $\text{Col}(\mathbf{A}_d)$? What are the projection $\mathbf{A}_d \hat{\mathbf{x}}$ and the squared error, and is $\hat{\mathbf{x}}$ unique? What does least squares reduce to for this matrix?